Foundations of AI 2: Deep Learning & Computer Vision

by

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Workshop 1 Recap

The Foundation: From Data to Intelligence

Key Concepts from Workshop 1

AI Learns from Data

Unlike traditional programming with explicit rules, AI discovers patterns from examples

Text → Numbers

Embeddings capture meaning: words with similar meanings get similar number representations

Attention & Transformers

AI focuses on relevant parts of input, enabling powerful language understanding

Foundation: AI uses data-driven learning + vector representations + attention to understand language

The AI Learning Loop

Key Insight

This same loop powers everything from image recognition to ChatGPT. The magic is in the architectures that define HOW the model processes data.

Workshop 2

Deep Learning & Computer Vision

From Neurons to Intelligent Systems

Today's Learning Journey

1

Neurons to Networks

How simple units combine to create complex intelligence

2

How AI Sees

CNNs and spatial intelligence for image understanding

3

How AI Remembers

LSTMs and sequential memory for time-based data

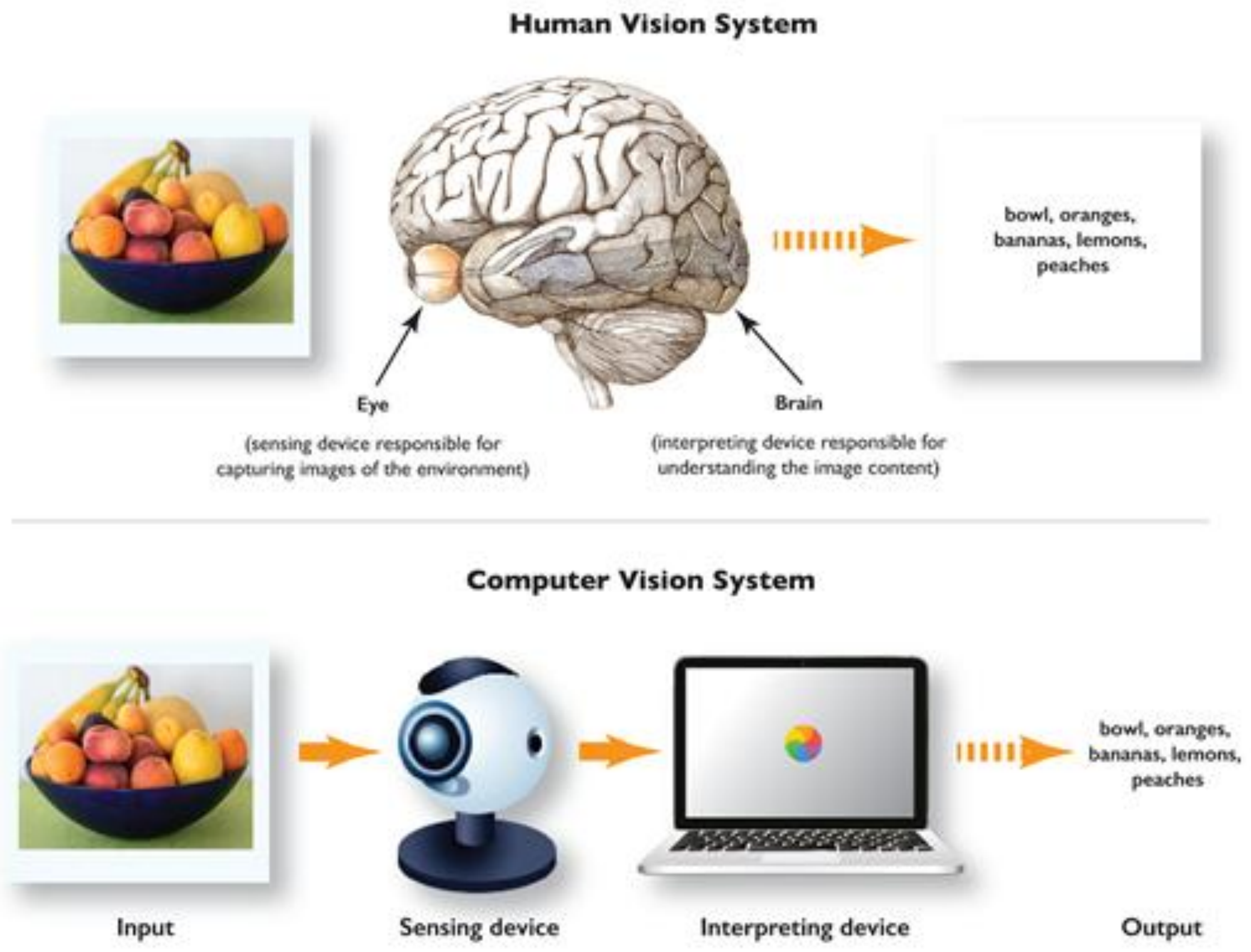
4

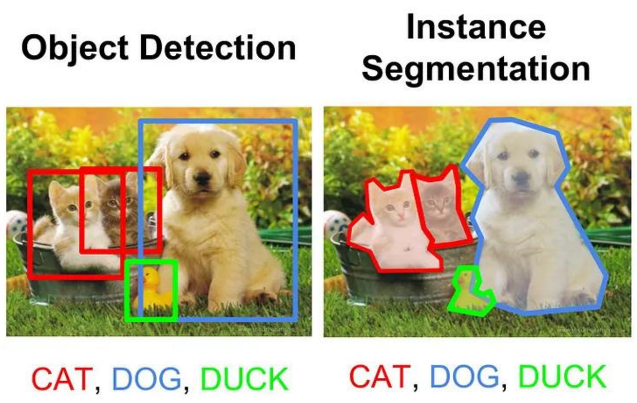
How AI Creates

Diffusion models that generate images from noise

Each notebook provides hands-on interactive visualizations to build your intuition!

What is Computer Vision?





1. Go to *jhub.dartmouth.edu*
2. Click *Start My Server*
3. Select *Reproducible Research Workshops*
4. *RR-workshops/foundations-of-ai*
5. Select *Notebook 0*
6. Click on *Kernel* and select *"Restart Kernel and Run All Cells"*

Workshop 2 Roundup

What We've Explored Together

Key Takeaways

- Neural networks are layers of simple units that learn complex patterns through training
- CNNs understand spatial relationships by scanning images with learned filters
- LSTMs have memory gates that decide what to remember and forget over time
- Diffusion models create images by learning to reverse the process of adding noise

The Big Picture

Raw Data

(Images, Text, etc.)

→

Neural Network

(Learning Engine)

→

Specialized Architecture

(CNN, LSTM, etc.)

→

Intelligent Output

(Recognition, Generation)

Workshop 1: Foundations

- How AI learns from data
- Text → Vector representations
- Attention & Transformers
- Language understanding

Workshop 2: Architectures

- Neural network fundamentals
- CNNs for spatial understanding
- LSTMs for sequential memory
- Diffusion for image generation

What's Next?

Workshop 3

GenAI Brainstorming

Model selection strategies and identifying AI bias in real applications

Workshop 4

Critical AI Literacy

Analyzing AI outputs and verification strategies for responsible use

Building AI intuition, one workshop at a time!

Thank you for your Attention!

Please let us know your thoughts:
dartgo.org/feedback